



Islamic University of Technology
Department of Computer Science and Engineering

Lab Test (Mid)

CSE 4404: Algorithms Lab
Summer 2024-25

Task A. Cloud Resource Scheduler

Time Limit: 1 second | Memory Limit: 1536 MB

You are designing an automated resource scheduler for a cloud hosting node with a maximum capacity of C GB of RAM. Given a set of N microservice containers, where each container i requires w_i GB of RAM and yields a revenue of r_i dollars, your goal is to select a subset of containers to deploy into the node to maximize the net profit (total revenue minus penalty). If S is the total RAM allocated, and the remaining unallocated RAM ($C - S$), where exceeds a threshold of E GB, a resource-waste penalty of \$5 per excess GB is deducted from the revenue. The penalty is calculated as:

$$\text{Penalty} = \begin{cases} 5 \times ((C - S) - E) & \text{if } (C - S) > E \\ 0 & \text{otherwise} \end{cases}$$

Your task is to find the maximum net profit (Revenue – Penalty) that can be achieved without exceeding the server capacity C .

Input Format

The input consists of three lines:

- The first line contains three space-separated integers: C (maximum capacity in GB), N (number of microservices), and E (threshold in GB).
- The second line contains N space-separated integers $w_1, w_2, \dots, w_N$, representing the RAM requirement of each microservice container in GB.
- The third line contains N space-separated integers $r_1, r_2, \dots, r_N$, representing the revenue generated by each microservice container in dollars.

Output Format

Print a single line containing a single integer representing the maximum net profit achievable.

Constraints

$$\begin{aligned} 1 &\leq C \leq 1000 \\ 1 &\leq N \leq 100 \\ 0 &\leq E \leq C \\ 1 &\leq w_i \leq C \\ 1 &\leq r_i \leq 1000 \end{aligned}$$

Examples

Sample Input	Sample Output
100 2 10 10 80 50 60	110
100 2 0 10 95 100 80	55

Explanation

Sample 1: In this case, $C = 100$, $N = 2$, and $E = 10$ with weights $\{10, 80\}$ and revenues $\{50, 60\}$.

- Selecting both containers: Total RAM $S = 10 + 80 = 90$ GB. Total Revenue = $50 + 60 = \$110$. Unallocated RAM = $100 - 90 = 10$ GB. Since $10 \leq E$ ($10 \leq 10$), Penalty = \$0. Net Profit = $110 - 0 = 110$.

Sample 2: In this case, $C = 100$, $N = 2$, and $E = 0$ with weights $\{10, 95\}$ and revenues $\{100, 80\}$.

- Selecting Container 1 ($w_1 = 10, r_1 = 100$): Revenue (100). Unallocated RAM = 90, Penalty = $5 \times 90 = \$450$. Net Profit = $100 - 450 = -350$.
- Selecting Container 2 ($w_2 = 95, r_2 = 80$): Revenue = \$80, Unallocated RAM = 5, Penalty = $5 \times 5 = \$25$. Net Profit = $80 - 25 = 55$.

Task B. Roads After the Storm

Time Limit: 2 seconds | Memory Limit: 256 MB

A big storm has flooded many roads in a region. There are n villages and m two way roads. One village, S , is the relief camp. Every other village gets its supplies from there. Before the storm, the relief team had already worked out $t(v)$, the shortest possible travel time from the camp S to each village v . Their whole supply plan depends on these numbers staying true.

Due to orders from a higher authority, the relief team can only keep exactly k of the m roads open tonight; the rest stay closed. Some villages may no longer be reachable from S in their original time $t(v)$, since a road they depended on could be among the ones that get closed.

A village v is called **safe** if, using only the k roads that stay open, the shortest travel time from S to v is still exactly $t(v)$. A village that becomes unreachable, or can only be reached in more time than before, is not safe.

The relief team gets to choose which k roads stay open. Your task is to help them choose these k roads by maximizing the number of safe villages, and report which villages end up safe.

Input Format

- The first line contains four integers n , m , k , and S . These are the number of villages, the total number of roads, the number of roads that can stay open, and the index of the relief camp village.
- Each of the next m lines contains three integers u , v , and w . These describe a two way road between village u and village v that takes w minutes to travel in either direction.

Output Format

Print two lines.

- The first line should contain a single integer, the number of safe villages.
- The second line should contain the indices of all safe villages, in ascending order, separated by spaces.

Note that village S is always safe, since its shortest travel time to itself is always 0 no matter which roads are kept open. So it will always appear in your output.

In case of a tie between two villages, the village with the lower travel time is picked to be safe.

Constraints

$$2 \leq n \leq 10^5$$

$$n - 1 \leq m \leq \min\left(2 \times 10^5, \frac{n(n-1)}{2}\right)$$

$$0 \leq k \leq m$$

$$1 \leq S \leq n$$

$$1 \leq w \leq 10^9$$

The road network is guaranteed to be connected.

No road connects a village to itself, and no two roads connect the same pair of villages.

It is guaranteed that $t(v)$ is distinct for every village v : no two villages have the same shortest travel time from S .

Examples

Sample Input	Sample Output
6 5 4 1 1 2 10 1 4 3 4 3 2 4 5 4 5 6 1	5 1 3 4 5 6
6 5 5 4 1 2 10 1 4 3 4 3 2 4 5 4 5 6 1	6 1 2 3 4 5 6
9 9 5 1 1 2 15 1 3 1 3 4 4 4 5 4 5 6 4 6 7 4 4 8 2 1 9 20 8 9 100	6 1 3 4 5 6 8

Explanation

Sample 1: The relief camp is village 1. With $k = 4$, villages 1, 3, 4, 5, and 6 are safe, while village 2 is not.

Sample 2: Same road network as Sample 1, but now the camp is village 4 and $k = 5$. All 6 villages are safe.

Sample 3: The relief camp is village 1. With $k = 5$, villages 1, 3, 4, 5, 6, and 8 are safe, while villages 2, 7, and 9 are not.

Task C. Minimum Pairing Cost

Time Limit: 1 second | Memory Limit: 256 MB

You are given $2n$ integers. You must divide all of them into exactly n pairs, so that every integer belongs to exactly one pair. The cost of a pair (x, y) is $|x - y|$. The total cost is the sum of the costs of all n pairs.

Your task is to find the minimum possible total cost.

Input Format

The input consists of two lines:

- The first line contains a single integer n , the number of pairs.
- The second line contains $2n$ space-separated integers $a_1, a_2, \dots, a_{2n}$.

Output Format

Print a single integer representing the minimum possible total cost.

Constraints

$$1 \leq n \leq 10^5$$
$$-10^9 \leq a_i \leq 10^9$$

Examples

Sample Input	Sample Output
3 8 1 4 10 2 7	6
2 -5 4 -2 9	8
4 10 1 8 3 8 3 1 10	0

Explanation

Sample 1: One possible division is $(4, 7)$, $(1, 2)$, and $(8, 10)$. The corresponding costs are 3, 1, and 2, so the total cost is 6. No other valid division produces a smaller total cost.

Sample 2: A minimum-cost division is $(-5, -2)$ and $(4, 9)$. Their costs are 3 and 5, giving a total cost of 8.

Sample 3: Each number can be paired with another number of the same value. Therefore, the cost of every pair is 0, and the minimum total cost is also 0.